GENERATIVE AI · RAG · PERSONAL PROJECT

AI Financial Research Assistant

A question-answering assistant that finds answers in long financial reports and shows where each answer came from.

Built using a practice dataset · Kasthuri Sathish

The problem

Analysts spend a lot of time skimming long financial reports to find specific facts. Generic chatbots can answer quickly, but they may invent figures and don't show their sources, so the answers are hard to trust.

What I built

A RAG application where a user asks a question in plain language and gets an answer drawn only from the documents, with a source reference to check it against.

How it works

Financial documents → Split into chunks → Embed with Sentence Transformers → Retrieve relevant chunks → Gemini / Bedrock with guardrails → Answer + source in Streamlit

Key product decisions

- Answer only from the retrieved documents, to reduce hallucinations.
- Say "Information not available in the source documents" when the answer isn't there.
- Show the source with every answer, so users can verify it.
- Used both Amazon Bedrock and the Gemini API to compare how each model handled complex financial tables and dense narrative text.

Testing and learnings

- Tested with 25 curated financial questions, checking each answer and its citation against the source documents.
- Refined chunk size and overlap so numerical tables stayed intact and relevant context wasn't cut off.
- Answer quality depends heavily on retrieval: if the right chunk isn't found, the answer can't be right.
- Limitation: built and tested on a practice dataset, not production data.

Tech stack

Python · Gemini API · Amazon Bedrock · Sentence Transformers · RAG · Streamlit

GitHub: github.com/kasthuri-tech/AI-financial-research-assistant- · Portfolio: kasthuri-tech.github.io